



WASTE MANAGEMENT

A growing need for smart solution

Presented by Fourth Group

INTRODUCTION

- **Improper waste disposal causes pollution, affects ecosystems, and poses serious health risks Effective.**
- **waste management helps conserve natural resources, reduce environmental damage, and promote sustainability.**
- **• Waste poses significant environmental and health challenges.**
- **• Traditional waste disposal methods are often inefficient and harmful.**
- **• Solution: A smart waste management system to streamline waste handling.Focuses on sustainability, automation, and reducing environmental impact.**

Working Principle

THE WASTE IS INITIALLY DROPPED INTO A BIN, WHERE A SERVO MOTOR IS USED TO RELEASE IT ONTO A CONVEYOR BELT. THE CONVEYOR BELT, DRIVEN BY A STEPPER MOTOR, MOVES THE WASTE FORWARD AND, AFTER A SET TIME DELAY, DROPS IT INTO AN INTERMEDIATE DETECTION BOX. INSIDE THIS BOX, SENSORS SUCH AS AN IR SENSOR, SOIL MOISTURE SENSOR, AND PROXIMITY SENSOR ARE USED TO IDENTIFY THE TYPE OF WASTE. BASED ON THE SENSOR READINGS, A SERVO MOTOR OPENS THE APPROPRIATE LID. SIMULTANEOUSLY, A ROTATING DISC POWERED BY A STEPPER MOTOR ALIGNS ITSELF TO DIRECT THE WASTE INTO THE CORRESPONDING FINAL BIN, ENSURING PROPER SEGREGATION.

Components

Sensors : IR Sensor, Inductive proximity sensor, Moisture Sensor

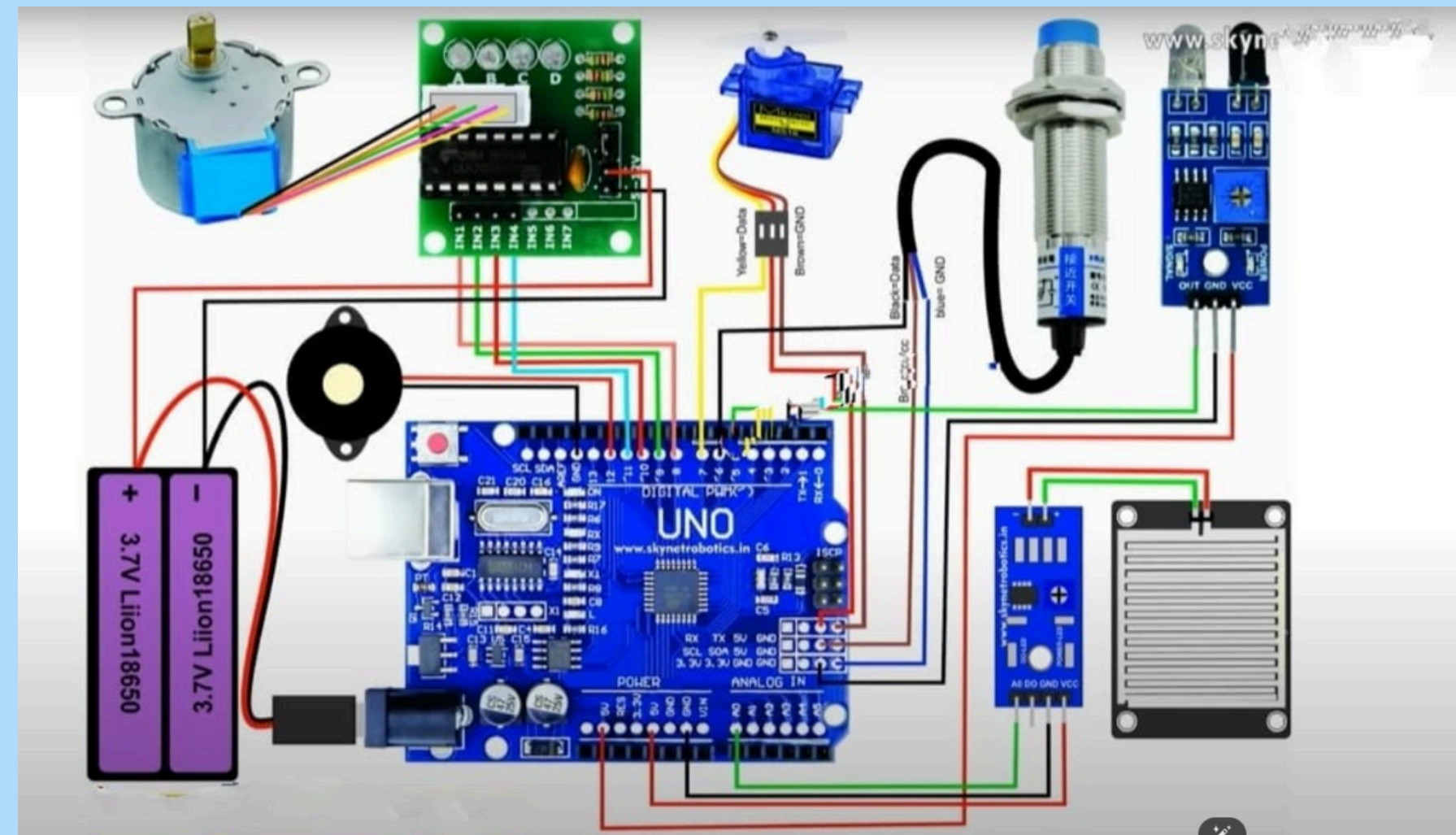
Micro Controller : Arduino, Bread Board

Motion Controller : Stepper Motor, Servo Motor, Wheel

Note : There is no external power supply

System Design and Connections

- Sensors collect data and send it to the micro controller
- Sensors can track waste pickup and get segregation tips.



Mechanism and Functioning

1. Waste Collection: Smart bins gather waste and monitor fill levels.
2. Waste Processing: Separated into recyclable, compostable, and landfill.

Advantages of Smart Waste Management Systems

- ✓ Reduces overflow and illegal dumping.**
- ✓ Saves fuel and labor through optimized routing.**
- ✓ Promotes recycling and environmental awareness.**
- ✓ Reduces landfill and pollution.**

Future Scope

Integration with smart cities and green policies.

- **Expansion into hazardous and e-waste management.**
- **Real-time air and soil pollution tracking from landfills.**

CONCLUSION

- * A smart waste management system ensures sustainable urban living.
 - * Reduces environmental harm while improving efficiency.
- * Technological integration is key to solving modern waste challenges.

THANKYOU!